

Qualification Pack



Assistant Lab Technician - Green Hydrogen

QP Code: HYC/Q4004

Version: 1.0

NSQF Level: 3.5

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HYC/Q4004: Assistant Lab Technician - Green Hydrogen

Brief Job Description

An Assistant Lab Technician - Green Hydrogen is responsible for receiving, labelling and analysing samples. The individual is responsible for designing and executing laboratory testing according standard procedures. Conducting experiments under defined conditions to verify/reject various types of hypotheses using refined scientific methods.

Personal Attributes

The individual should have ability to give attention to detail, technical aptitude, adaptability, teamwork and a strong commitment to safety and sustainability. The person should have the ability to work individually / team round the clock and must always promote safety.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4001: Hydrogen Safety, Security and Health Management Procedures](#)
2. [HYC/N4008: Green Hydrogen Industry and Laboratory regulations](#)
3. [HYC/N4009: Experimentation and Analysis in Green Hydrogen](#)
4. [HYC/N9301: Working effectively in a team](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
Country	India
NSQF Level	3.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO/2015 3116.0500

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Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th (Petroleum/chemical) OR Pursuing 3rd year of 3-year diploma after 10th (Petroleum/chemical) OR 12th grade pass with 2 year NTC/ CITS/NAC OR 10th grade pass with 2 year NTC plus 1 year NAC OR 12th grade Pass (Science) with 1 Year of experience OR 10th grade pass plus 1-year NTC/ NAC with 1 Year of experience OR Previous relevant Qualification of NSQF Level (3) with 1-2 Years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/11/2026
NSQC Approval Date	30/11/2023
Version	1.0
Reference code on NQR	NCVET- QG-3.5-HY-01365-2023-V1-HSSCI
NQR Version	1.0

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HYC/N4001: Hydrogen Safety, Security and Health Management Procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace where Hydrogen is produced and handled.

Scope

The scope covers the following :

- Understanding Properties of Hydrogen from safety point of view
- Safe handling of Hydrogen
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.
- PC2.** Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions
- PC3.** Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.
- PC4.** Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.
- PC5.** Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire
- PC6.** capability to accurately read and interpret data from sensing equipment used in hydrogen production processes
- PC7.** apply key safety parameters for handling hydrogen & oxygen
- PC8.** Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames
- PC9.** Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards
- PC10.** competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace
- PC11.** Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations
- PC12.** Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.

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- PC13.** safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries
- PC14.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC15.** Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.
- PC16.** Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.
- PC17.** Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.

Ensure the following fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC18.** Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly
- PC19.** Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard
- PC20.** Promote and enforce following good housekeeping practice in order to prevent fire hazards
- PC21.** Regularly inspect work areas for compliance with housekeeping standards
- PC22.** identify and document issues concerning the hydrogen safety in work place
- PC23.** Collaborate with relevant departments to address and rectify safety concerns promptly
- PC24.** Ensure that work areas and escape routes are well maintained and free from any hindrance
- PC25.** Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation
- PC26.** Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard
- PC27.** Ensure that personnel are trained in preparing comprehensive accident/incident reports
- PC28.** Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC29.** provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues
- PC30.** Be proficient in basic techniques of bandaging
- PC31.** Lead by example in responding promptly and appropriately to an accident
- PC32.** perform rescue activity during an accident in real or simulated environments
- PC33.** Familiarize personnel with correct escape routes during an emergency
- PC34.** demonstrate correct method to rescue injured people and others during an emergency
- PC35.** Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance
- KU12.** the concept of Mission Karmayogi
- KU13.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen</i>	17	27	-	-
PC1. Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.	1	2	-	-
PC2. Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions	1	1	-	-
PC3. Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.	1	2	-	-
PC4. Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.	1	2	-	-
PC5. Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire	1	2	-	-
PC6. capability to accurately read and interpret data from sensing equipment used in hydrogen production processes	1	2	-	-
PC7. apply key safety parameters for handling hydrogen & oxygen	1	2	-	-
PC8. Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames	1	1	-	-
PC9. Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards	1	1	-	-
PC10. competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations	1	1	-	-
PC12. Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.	1	1	-	-
PC13. safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries	1	2	-	-
PC14. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC15. Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.	1	2	-	-
PC16. Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.	1	1	-	-
PC17. Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.	1	1	-	-
<i>Ensure the following fire safety procedures</i>	11	19	-	-
PC18. Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly	1	2	-	-
PC19. Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard	1	2	-	-
PC20. Promote and enforce following good housekeeping practice in order to prevent fire hazards	1	2	-	-
PC21. Regularly inspect work areas for compliance with housekeeping standards	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify and document issues concerning the hydrogen safety in work place	1	2	-	-
PC23. Collaborate with relevant departments to address and rectify safety concerns promptly	1	2	-	-
PC24. Ensure that work areas and escape routes are well maintained and free from any hindrance	1	2	-	-
PC25. Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation	1	2	-	-
PC26. Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard	1	1	-	-
PC27. Ensure that personnel are trained in preparing comprehensive accident/incident reports	1	1	-	-
PC28. Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions	1	1	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	7	14	-	-
PC29. provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues	1	2	-	-
PC30. Be proficient in basic techniques of bandaging	1	2	-	-
PC31. Lead by example in responding promptly and appropriately to an accident	1	2	-	-
PC32. perform rescue activity during an accident in real or simulated environments	1	2	-	-
PC33. Familiarize personnel with correct escape routes during an emergency	1	2	-	-
PC34. demonstrate correct method to rescue injured people and others during an emergency	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.	1	2	-	-
NOS Total	35	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4001
NOS Name	Hydrogen Safety, Security and Health Management Procedures
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4008: Green Hydrogen Industry and Laboratory regulations

Description

This unit covers standard testing procedures and protocols for all the raw material and final products in a Green Hydrogen Quality Control laboratory. This includes procedures and practices that candidates need to follow to ensure quality control and quality assurance in the production of green hydrogen in a healthy, safe and secure work environment.

Scope

The scope covers the following :

- Understanding Green Hydrogen Industry and Laboratory regulations.

Elements and Performance Criteria

Understanding Green Hydrogen Industry and Laboratory regulations

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the fundamental concept of green hydrogen, which is hydrogen produced using renewable energy sources and its importance in the context of sustainable energy.
- PC2.** Understand various methods or pathways for producing green hydrogen, such as electrolysis, biomass conversion and solar-driven processes.
- PC3.** Recognize the critical role of raw materials, particularly water and renewable energy sources in the production of green hydrogen.
- PC4.** Have awareness of the quality standards and criteria that govern the characteristics of green hydrogen, including purity levels, impurity limits and environmental considerations.
- PC5.** Have knowledge of the specific rules and regulations governing quality control laboratories, encompassing safety protocols, quality assurance procedures and compliance with industry standards.
- PC6.** Understand the regulatory requirements specific to research and development labs, including considerations related to intellectual property, data management and innovation protection.
- PC7.** Understand the duties and responsibilities of a lab technician, which may include sample analysis, instrument maintenance, data recording, adherence to safety protocols and collaboration with other teams.
- PC8.** Carry knowledge of the essential functions that laboratories serve within a green hydrogen production plant, such as quality control, process optimization, troubleshooting and supporting research and development efforts.
- PC9.** Understand implementation of principles and procedures that ensure the integrity, reliability and reproducibility of laboratory data.
- PC10.** Have knowledge of the quality management and control measures applied in manufacturing processes to ensure product safety and consistency.
- PC11.** Carry awareness of the importance of accurate and organized documentation of laboratory procedures, data and results.

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- PC12.** Understand safety protocols, risk assessment, emergency procedures and the proper handling of hazardous materials and equipment to maintain a safe laboratory environment.
- PC13.** Understand the concepts of data repeatability (consistency of results within a single test) and reproducibility (consistency of results across different tests or operators).
- PC14.** Use competence in using statistical methods to analyze and interpret laboratory data, including measures of central tendency, variation and hypothesis testing.
- PC15.** Carry ability to effectively communicate laboratory findings through clear and accurate reporting, including documentation of methods, results and conclusions.
- PC16.** Show proficiency in utilizing laboratory information management systems for data storage, retrieval, organization and overall lab management in quality control (QC) or research and development (R&D) environments.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of a quality control lab and the technician in a production unit
- KU2.** Basic analytical skills and a knowledge about different techniques and their importance
- KU3.** Knowledge of industry regulations and standards,
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance
- KU12.** the concept of Mission Karmayogi
- KU13.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Basic computer operations, Interpret data, Data reporting, knowledge of basic statistics
- GS2.** communicate by writing
- GS3.** read standard test methods, SOPs, instructions, guidelines/procedures and reports
- GS4.** communicate effectively and share the information efficiently
- GS5.** identify and report potential sources of error and their rectification
- GS6.** how to plan the work to meet the deadline
- GS7.** working under tight time deadlines with high production volumes
- GS8.** apply problem solving approaches in different situations

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GS9. apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Green Hydrogen Industry and Laboratory regulations</i>	20	30	-	-
PC1. Understand the fundamental concept of green hydrogen, which is hydrogen produced using renewable energy sources and its importance in the context of sustainable energy.	1	1	-	-
PC2. Understand various methods or pathways for producing green hydrogen, such as electrolysis, biomass conversion and solar-driven processes.	2	3	-	-
PC3. Recognize the critical role of raw materials, particularly water and renewable energy sources in the production of green hydrogen.	2	3	-	-
PC4. Have awareness of the quality standards and criteria that govern the characteristics of green hydrogen, including purity levels, impurity limits and environmental considerations.	1	1	-	-
PC5. Have knowledge of the specific rules and regulations governing quality control laboratories, encompassing safety protocols, quality assurance procedures and compliance with industry standards.	1	2	-	-
PC6. Understand the regulatory requirements specific to research and development labs, including considerations related to intellectual property, data management and innovation protection.	1	1	-	-
PC7. Understand the duties and responsibilities of a lab technician, which may include sample analysis, instrument maintenance, data recording, adherence to safety protocols and collaboration with other teams.	1	1	-	-
PC8. Carry knowledge of the essential functions that laboratories serve within a green hydrogen production plant, such as quality control, process optimization, troubleshooting and supporting research and development efforts.	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Understand implementation of principles and procedures that ensure the integrity, reliability and reproducibility of laboratory data.	1	2	-	-
PC10. Have knowledge of the quality management and control measures applied in manufacturing processes to ensure product safety and consistency.	1	2	-	-
PC11. Carry awareness of the importance of accurate and organized documentation of laboratory procedures, data and results.	1	2	-	-
PC12. Understand safety protocols, risk assessment, emergency procedures and the proper handling of hazardous materials and equipment to maintain a safe laboratory environment.	2	2	-	-
PC13. Understand the concepts of data repeatability (consistency of results within a single test) and reproducibility (consistency of results across different tests or operators).	1	2	-	-
PC14. Use competence in using statistical methods to analyze and interpret laboratory data, including measures of central tendency, variation and hypothesis testing.	1	2	-	-
PC15. Carry ability to effectively communicate laboratory findings through clear and accurate reporting, including documentation of methods, results and conclusions.	1	2	-	-
PC16. Show proficiency in utilizing laboratory information management systems for data storage, retrieval, organization and overall lab management in quality control (QC) or research and development (R&D) environments.	1	2	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4008
NOS Name	Green Hydrogen Industry and Laboratory regulations
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	3.5
Credits	4.5
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

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HYC/N4009: Experimentation and Analysis in Green Hydrogen

Description

This unit covers standard testing procedures and protocols for all the raw material and final products in a Green Hydrogen Quality Control laboratory. This includes procedures and practices that candidates need to follow to ensure quality control and quality assurance in the production of green hydrogen in a healthy, safe and secure work environment.

Scope

The scope covers the following :

- Specifications of water and green hydrogen for various applications
- Impact of different types of impurities on different types of Electrolyzers
- Standard Test methods for water analysis, biomass testing, oxygen and green hydrogen testing, equipment Calibration, basic upkeep of equipment.

Elements and Performance Criteria

Experimentation and Analysis

To be competent, the user/individual on the job must be able to:

- PC1.** Understand fundamental principles and concepts behind various analytical techniques commonly used in QA/QC processes. determine types of electrolyzers, its function and operating principles
- PC2.** Show proficiency in applying these techniques to assess the quality, purity and characteristics of green hydrogen and related materials
- PC3.** Carry knowledge of the quality parameters, purity levels and other specifications required for water and green hydrogen to meet the demands of various industrial and commercial applications.
- PC4.** Recognition of the various sources and types of impurities that can affect the quality of water used in green hydrogen production
- PC5.** Understand the ASTM (American Society for Testing and Materials) standards for Type-II water, which typically sets the quality standard for laboratory-grade water.
- PC6.** Understand the different impurity types, such as dissolved gases, ions and particulates and how they can impact the efficiency and performance of different electrolyzer technologies.
- PC7.** Have knowledge and skills related to electrochemical methods used in analysing and evaluating electrochemical processes in green hydrogen production
- PC8.** Understand techniques for characterizing materials used in hydrogen production systems, such as catalysts and membranes.
- PC9.** Show proficiency in conducting chemical analyses of both liquid and gas phases to assess the composition and quality of hydrogen and related materials.
- PC10.** Understand the methods and techniques for characterizing biomass feedstocks, including composition analysis and moisture content determination.
- PC11.** Recognize the impurities that can be generated or introduced during the conversion of biomass to hydrogen and their potential impact on the process.

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- PC12.** Understand established and recognized test methods and procedures for analysing water quality, biomass composition, oxygen content and green hydrogen purity.
- PC13.** Have knowledge and skills for calibrating laboratory equipment and instruments to ensure accuracy and reliability of measurements.
- PC14.** perform routine maintenance tasks to keep laboratory equipment in good working condition, ensuring accurate and consistent results.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of a quality control lab and the technician in a production unit
- KU2.** Basic analytical skills and a knowledge about different techniques and their importance
- KU3.** Knowledge of industry regulations and standards
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance
- KU12.** the concept of Mission Karmayogi
- KU13.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Basic computer operations, Interpret data, Data reporting, knowledge of basic statistics
- GS2.** communicate by writing
- GS3.** read standard test methods, SOPs, instructions, guidelines/procedures and reports
- GS4.** communicate effectively and share the information efficiently
- GS5.** identify and report potential sources of error and their rectification
- GS6.** how to plan the work to meet the deadline
- GS7.** working under tight time deadlines with high production volumes
- GS8.** apply problem solving approaches in different situations
- GS9.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Experimentation and Analysis</i>	20	30	-	-
PC1. Understand fundamental principles and concepts behind various analytical techniques commonly used in QA/QC processes. determine types of electrolyzers, its function and operating principles	1	2	-	-
PC2. Show proficiency in applying these techniques to assess the quality, purity and characteristics of green hydrogen and related materials	1	2	-	-
PC3. Carry knowledge of the quality parameters, purity levels and other specifications required for water and green hydrogen to meet the demands of various industrial and commercial applications.	1	2	-	-
PC4. Recognition of the various sources and types of impurities that can affect the quality of water used in green hydrogen production	1	2	-	-
PC5. Understand the ASTM (American Society for Testing and Materials) standards for Type-II water, which typically sets the quality standard for laboratory-grade water.	2	3	-	-
PC6. Understand the different impurity types, such as dissolved gases, ions and particulates and how they can impact the efficiency and performance of different electrolyzer technologies.	2	3	-	-
PC7. Have knowledge and skills related to electrochemical methods used in analysing and evaluating electrochemical processes in green hydrogen production	1	1	-	-
PC8. Understand techniques for characterizing materials used in hydrogen production systems, such as catalysts and membranes.	2	2	-	-
PC9. Show proficiency in conducting chemical analyses of both liquid and gas phases to assess the composition and quality of hydrogen and related materials.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Understand the methods and techniques for characterizing biomass feedstocks, including composition analysis and moisture content determination.	2	3	-	-
PC11. Recognize the impurities that can be generated or introduced during the conversion of biomass to hydrogen and their potential impact on the process.	2	2	-	-
PC12. Understand established and recognized test methods and procedures for analysing water quality, biomass composition, oxygen content and green hydrogen purity.	1	2	-	-
PC13. Have knowledge and skills for calibrating laboratory equipment and instruments to ensure accuracy and reliability of measurements.	1	1	-	-
PC14. perform routine maintenance tasks to keep laboratory equipment in good working condition, ensuring accurate and consistent results.	1	2	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4009
NOS Name	Experimentation and Analysis in Green Hydrogen
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	3.5
Credits	5
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

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HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4.5
Credits	1.5
Version	6.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 50% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 50

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4001.Hydrogen Safety, Security and Health Management Procedures	35	60	-	-	95	30
HYC/N4008.Green Hydrogen Industry and Laboratory regulations	20	30	-	-	50	30
HYC/N4009.Experimentation and Analysis in Green Hydrogen	20	30	-	-	50	20
HYC/N9301.Working effectively in a team	20	30	-	-	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	115	180	-	-	295	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

Qualification Pack

NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.